Aircraft Trajectory Reconstruction

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GitHub: github.com/Joe-H04/ml-aeronautics

Docker: https://hub.docker.com/r/20404/ml-aeronautics

Data Source

Historical ADS-B/ADS-C flight tracks are ingested from the **OpenSky Network** Trino table `flights_data4`, filtered to flights departing from or arriving at ten major European airports (EDDF, EGLL, LFPG, EHAM, LEMD, LIRF, LOWW, LSZH, EKCH, EDDM). Each day's `track` array is unnested into point-level rows, cleaned (sort by time, drop duplicates and NaNs, remove on-ground and high-altitude points above 15 000 m), and written as per-flight parquet files. A CRC32 hash on `icao24|firstseen` gives a deterministic 80/20 train/test split. The current workspace contains **7 215** training trajectories and **1 808** held-out test trajectories. The FastAPI service also queries the live OpenSky REST API for real-time reconstructions.

Approach

The task is to reconstruct missing segments inside an aircraft trajectory given context on both sides of the gap. We combine a strong geometric baseline with a learned residual correction. A fixed-length context window ($C = 20$ points before and after the gap) is used to predict $G = 10$ hidden points. A linear interpolation is first drawn between the last pre-gap point and the first post-gap point, and a bidirectional LSTM then predicts a *residual correction* on top of that baseline. Input features are anchor-centred latitude/longitude offsets, altitude offset, and inter-point time deltas (scaled). The final output layer is zero-initialised, so the untrained network reduces exactly to the baseline interpolation.

Methods

- **Great-circle baseline:** haversine distance with spherical linear interpolation on the sphere, linear altitude.
- **Linear baseline:** straight-line interpolation in lat/lon/altitude space.
- **Kalman filter + RTS smoother:** constant-velocity 6-D state (position + velocity), bidirectional smoothing pass used for classical comparison.
- **Residual BiLSTM (main model):** input $(40, 4)$, BiLSTM-64 (return sequences) $\rightarrow$ Dropout 0.2 $\rightarrow$ BiLSTM-32 $\rightarrow$ Dropout 0.2 $\rightarrow$ Dense-64 (ReLU) $\rightarrow$ Dense-30 $\rightarrow$ Reshape $(10, 3)$. Trained with Adam + Huber loss ($\delta = 1$), 50 epochs, batch 128, early stopping (patience 6), seed 42.
- **Serving:** FastAPI exposes `/reconstruct`, `/demo`, and live `/opensky/{icao24}` endpoints, with a Leaflet + globe.gl browser UI. The whole stack (model, API, data) is packaged via `Dockerfile` and `docker-compose.yml` for reproducible deployment.
- **Emissions sensitivity:** route-distance error is converted to a CO_2 proxy using 9.48 kg CO_2/km (fuel 3.0 kg/km, CI 3.16).

Results

Evaluation was run on **5 418** held-out gaps (3 random windows per eligible test flight). Per-gap errors use the haversine great-circle distance.

Method	Mean (km)	Median (km)	Mean p95 (km)	Wins / 5418
Residual BiLSTM	**18.11**	**2.21**	44.23	**2 725**
Great-circle baseline	22.49	2.94	**40.70**	1 415
Linear baseline	22.78	2.95	41.09	1 278

The residual BiLSTM reduces mean reconstruction error by $(22.49 - 18.11)/22.49 \approx \mathbf{19.4}\%$ relative to the great-circle baseline and wins the per-gap comparison on roughly half of the test set ($2725/5418 \approx 50.3\%$), versus 26.1% for great-circle and 23.6% for linear. The median error drops from 2.94 km to 2.21 km, indicating that typical-case reconstructions improve markedly. The p95 error remains close to the baselines, which shows the model still produces a small number of heavy-tail outliers on particularly sparse or unusual flight segments — a known target for future work (variable gap length, richer features, uncertainty estimates).